Variation and selection

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1 Core Variation [1]

Which comparison is an example of variation within a species?

A the difference between a fish and a flowering plant

B different body lengths among adult individuals of one fish species

C the difference between an animal cell and a bacterial cell

D the difference between a young fish and the adult it grows into

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2 · Part 1 Core Variation [1]

Which measurement is most likely to show continuous variation in a population?

A ABO blood group

B whether pea seeds are smooth or wrinkled

C whether a plant has one of two inherited flower colours

D adult body mass

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3 · Part 1 Core Variation [1]

Which observation is evidence of discontinuous variation?

A individuals fall into distinct groups with no intermediate phenotypes

B individuals have every possible length within a range

C all masses are arranged along a continuous scale

D values can be measured to more decimal places

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4 · Part 1 Core Variation [1]

Identical plant genotypes grown with different nutrient supplies reach different heights. Which conclusion is supported?

A All variation must be caused by mutation.

B Height can never be inherited.

C The environment can contribute to continuous variation.

D Height is an example of discontinuous variation.

5 · Part 1 Core Variation [1]

What is the original source of a new allele?

A meiosis

B random fertilisation

C selective breeding

D mutation

6 · Part 1 Core Variation [1]

Which exposure can increase mutation rate?

A a lower temperature

B ionising radiation

C a higher oxygen concentration

D a diet high in protein

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7 · Part 1 Extended Variation [1]

Which event is a gene mutation?

A the separation of chromosomes during meiosis

B the random fusion of gametes at fertilisation

C a random change in a DNA base sequence

D the crossing of two selected parents by a breeder

8 · Part 1 Extended Variation [1]

Which set contains only sources of genetic variation?

A mutation, meiosis and random fertilisation

B mutation, diet and exercise

C meiosis, nutrition and climate

D random fertilisation, temperature and light intensity

9 · Part 1 Core Adaptive features [1]

Which feature best fits the definition of an adaptive feature?

A an injury acquired during life

B an inherited feature increasing survival and reproduction in an environment

C a feature learned during an organism's life

D any inherited feature, whether or not it is useful

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10 · Part 1 Core Adaptive features [1]

A nocturnal desert mammal avoids daytime heat. What is the most likely benefit of this behaviour?

A it no longer needs to drink water

B it avoids competition with every other species

C it is less exposed to high temperatures and loses less water

D it obtains more energy from each meal

11 · Part 1 Extended Adaptive features [1]

Which feature is most useful for reducing transpiration in a xerophyte?

A a thin cuticle

B a large number of stomata on the upper leaf surface

C large, flat leaves with a large surface area

D a thick waxy cuticle

12 · Part 1 Core Selection [1]

During natural selection, which individuals are most likely to contribute alleles to the next generation?

A those that survive and reproduce more successfully in that environment

B those that need a new feature most urgently

C those with the largest body size

D those chosen by humans for breeding

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13 · Part 1 Core Selection [1]

Which action is selective breeding?

A allowing every animal to mate without selection

B treating all animals with growth hormones to increase yield

C crossing animals chosen for high milk yield and selecting high-yielding offspring

D changing an animal's genes by inserting a bacterial gene

14 · Part 1 Extended Selection [1]

A bacterial population becomes resistant after repeated antibiotic treatment. Which explanation is best?

A resistant variants survive treatment and reproduce

B each bacterium chooses to become resistant

C the antibiotic causes all bacteria to mutate and become resistant

D all susceptible bacteria survive better than resistant ones

15 · Part 1 Extended Selection [1]

What distinguishes artificial selection from natural selection?

A Artificial selection produces no inherited changes.

B Humans choose which individuals reproduce in artificial selection.

C Natural selection acts on acquired features rather than inherited ones.

D Only artificial selection involves variation.

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16 · Part 1 Core Variation [4]

A class measures the hand spans of adult volunteers from one population.

(a) Explain why these measurements are an example of variation and identify the type. [2]

(b) State two influences that can contribute to differences in this type of feature. [2]

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17 · Part 1 Core Variation [4]

A survey assigns people to blood groups A, B, AB and O.

(a) Explain why these results show discontinuous variation. [2]

(b) Describe the usual cause of discontinuous variation and compare this with the cause of continuous variation. [2]

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18 · Part 1 Core Variation [4]

A student measures 40 leaves. Lengths range from 18 mm to 61 mm.

(a) Calculate the range and identify the type of variation. [2]

(b) The student wants to compare leaf length between two plants. State two steps that make this comparison fair and reliable. [2]

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19 · Part 1 Core Variation [4]

A population acquires a new allele after a genetic change.

(a) Describe mutation and its role in forming alleles. [2]

(b) State two types of factor that can increase mutation rate. [2]

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20 · Part 1 Extended Variation [4]

Two siblings inherit different combinations of their parents' alleles.

(a) Describe a gene mutation. [2]

(b) State two processes, other than mutation, that contribute to genetic variation between siblings. [2]

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21 · Part 1 Core Variation [4]

Two groups of genetically identical seedlings are grown under different light intensities and develop different masses.

(a) Explain why their difference in mass does not prove a difference in genes. [2]

(b) Describe two controls needed when testing the effect of light alone. [2]

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22 · Part 1 Core Variation [4]

Researchers record body length and seed shape in a pea population. Seed shape is either smooth or wrinkled.

(a) Describe how the two sets of phenotypes differ. [2]

(b) Describe how each type of data should be collected and recorded. [2]

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23 · Part 1 Extended Variation [4]

A species reproduces sexually. A new DNA sequence appears in one lineage.

(a) Explain which process can create the new sequence and describe that process. [2]

(b) Explain how sexual reproduction can increase variation without creating a new allele. [2]

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24 · Part 1 Extended Adaptive features [4]

A desert plant has rolled leaves and stomata in sunken pits.

(a) Explain how these two features reduce water loss. [2]

(b) Explain the benefit of a thick cuticle and water-storage tissue. [2]

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25 · Part 1 Core Adaptive features [4]

A water bird has broad webbed feet and a waterproof feather covering.

(a) Explain how each feature may aid survival in an aquatic environment. [2]

(b) Describe two requirements for either feature to be called an adaptive feature. [2]

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26 · Part 1 Core Selection [4]

A beetle population contains inherited light and dark forms. Dark beetles are harder for predators to see on newly darkened ground.

(a) Explain why dark beetles may leave more offspring. [2]

(b) Explain how the frequency of dark beetles can change over generations. [2]

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27 · Part 1 Core Selection [4]

A breeder wants goats that produce more milk.

(a) Describe the first two stages of a selective-breeding programme. [2]

(b) Describe how the programme should continue. [2]

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28 · Part 1 Extended Selection [4]

Before treatment, a small fraction of bacteria resist an antibiotic.

(a) Explain the immediate effect of antibiotic treatment on the population. [2]

(b) Explain why the resistant fraction may increase after treatment. [2]

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29 · Part 1 Extended Selection [4]

A wild grass population survives repeated droughts. A farmer separately breeds grass selected for high seed yield.

(a) Explain how the wild population can become adapted to drought. [2]

(b) Compare how reproductive success is determined in the two populations. [2]

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30 · Part 1 Core Selection [4]

A fish species produces many more young than can survive in a pond.

(a) Explain why this can lead to a struggle for survival. [2]

(b) Explain why survival in this struggle depends on inherited variation rather than on choices made by individual fish. [2]

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